Practical - 3

Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

A. Theory

1. How does the layout adapt for mobile, tablet, and desktop screens?

Answer:
The StudentHub layout is designed using a mobile-first approach. The basic layout is created for smaller screens first, and then CSS media queries are used to adjust the layout for tablets and desktops.

On mobile screens, content is arranged mostly in a single column so that it remains easy to read. On tablets, elements can be arranged into multiple columns where space is available. On desktop screens, wider layouts, grids, and multiple content sections can be displayed.

This approach allows the same website to work properly on different screen sizes without creating separate websites for each device.

Reference:
MDN Web Docs – Responsive design
https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design

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2. Which layout technique is used and why?

Answer:
Both CSS Flexbox and CSS Grid are used in the StudentHub project.

Flexbox is mainly used for one-dimensional layouts such as navigation bars, buttons, cards, and arranging elements horizontally or vertically.

CSS Grid is used when a two-dimensional layout is required, such as displaying multiple cards, dashboard sections, events, or other content in rows and columns.

Using both techniques makes the layout easier to organize and responsive across different screen sizes.

Reference:
MDN Web Docs – CSS Flexbox
https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Flexbox

MDN Web Docs – CSS Grid
https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Grids

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3. Are typography, spacing, colors, and contrast readable?

Answer:
Yes. Typography, spacing, colors, and contrast are selected to make the StudentHub interface easy to read and navigate.

Readable font sizes are used for headings and normal text. Proper spacing is maintained between sections, cards, buttons, and form elements. The color combination provides sufficient contrast between the background and text. Important buttons and interactive elements are also visually distinguishable.

The same design system is maintained across the different pages to provide a consistent user experience.

Reference:
MDN Web Docs – CSS styling
https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Styling_basics

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4. Is the framework used consistently without unnecessary inline CSS?

Answer:
Yes. If Bootstrap is used in the StudentHub project, its components and utility classes are used consistently wherever they are useful.

Custom CSS is used for project-specific styling that is not provided by the framework. Unnecessary inline CSS is avoided because keeping styles in CSS files makes the project easier to maintain and update.

The framework and custom CSS are organized so that the design remains consistent across all pages.

Reference:
Bootstrap Documentation
https://getbootstrap.com/docs/

B. Implementation/ Output/Result

The StudentHub portal was designed using **HTML5, CSS3, Flexbox, CSS Grid, and responsive media queries**.

Responsive layouts were implemented for the following pages:

1. Home
2. About
3. Registration
4. Dashboard
5. Events

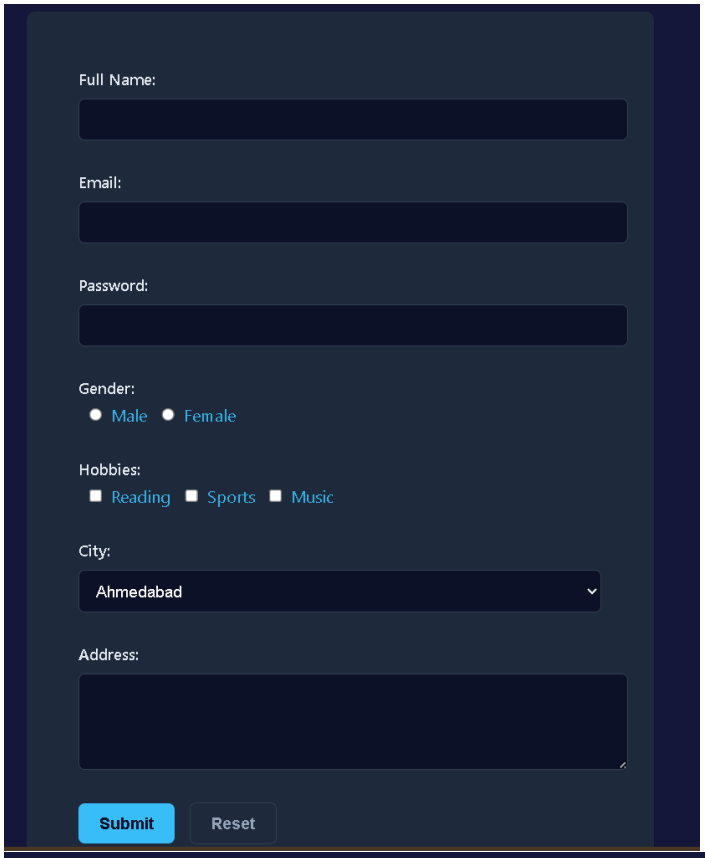
Example Responsive CSS

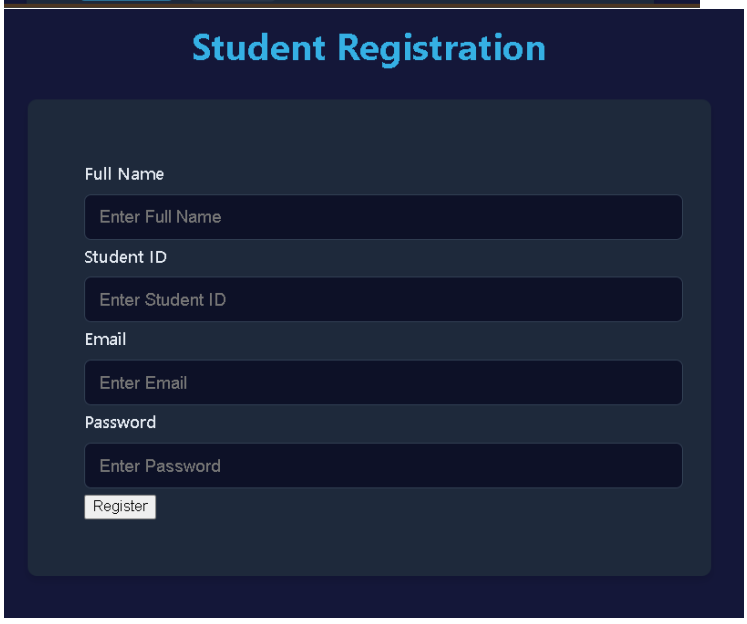
Result

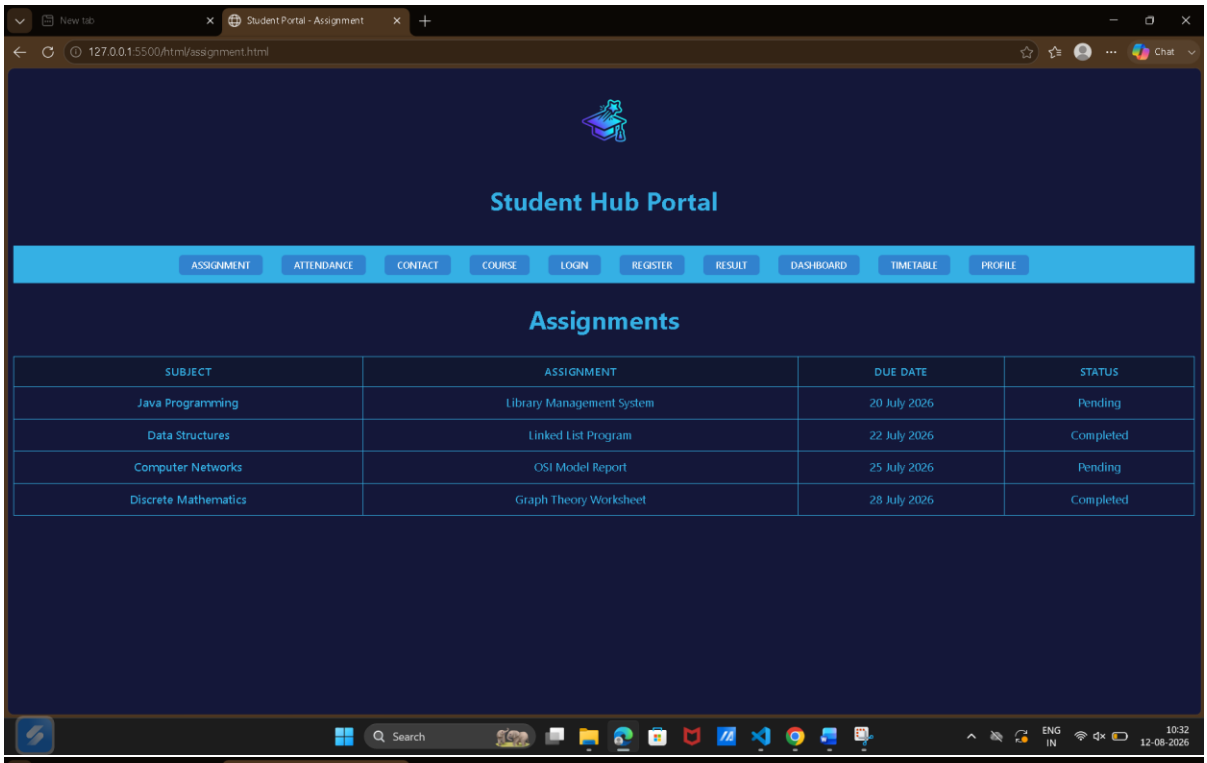
The StudentHub pages were successfully designed with responsive layouts. CSS Grid was used for structured content layouts, while Flexbox was used for navigation and one-dimensional arrangements.

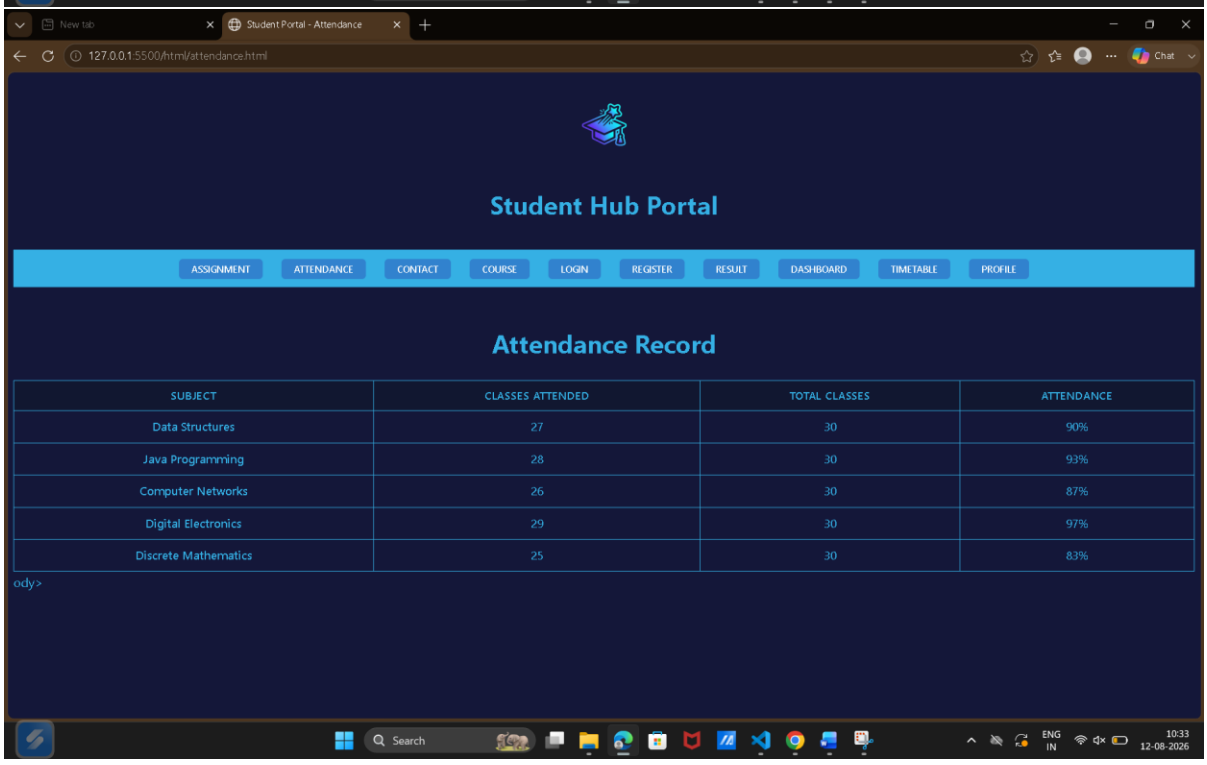
The interface adapts to mobile, tablet, and desktop screen sizes while maintaining consistent typography, spacing, colors, and overall design.

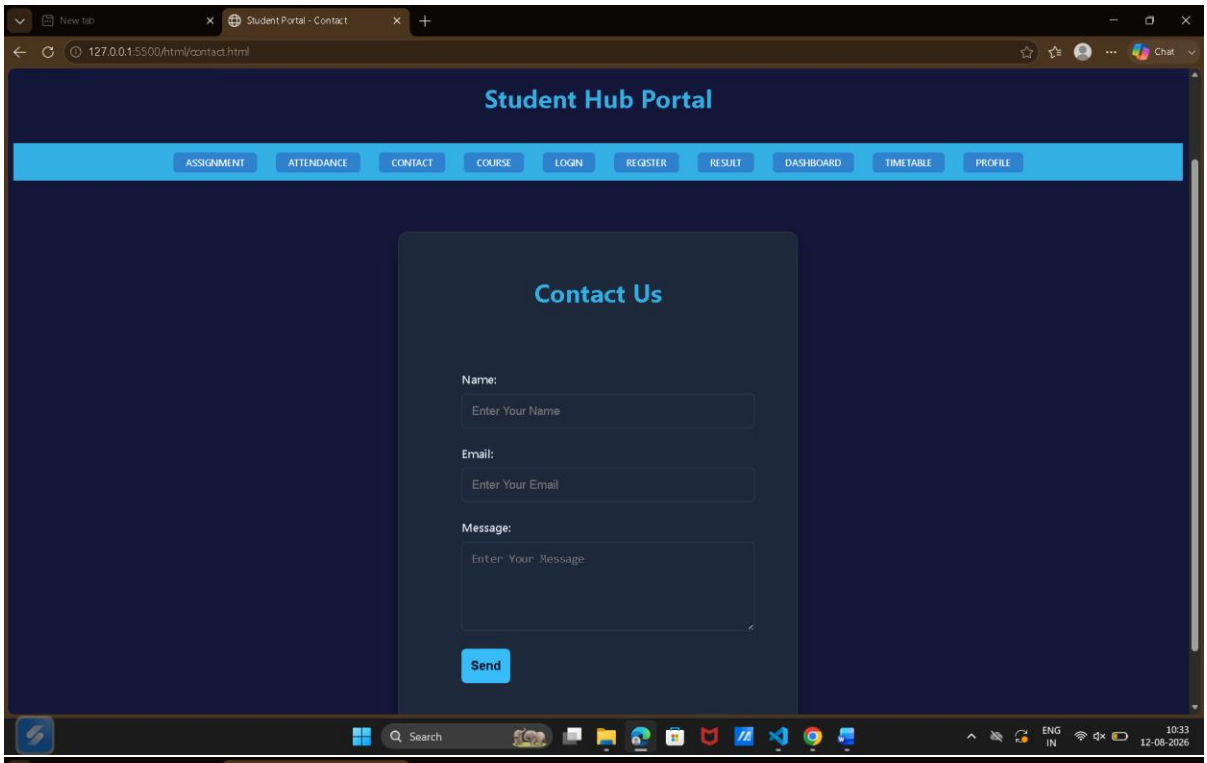
C. Evidence of Work (Screenshots)

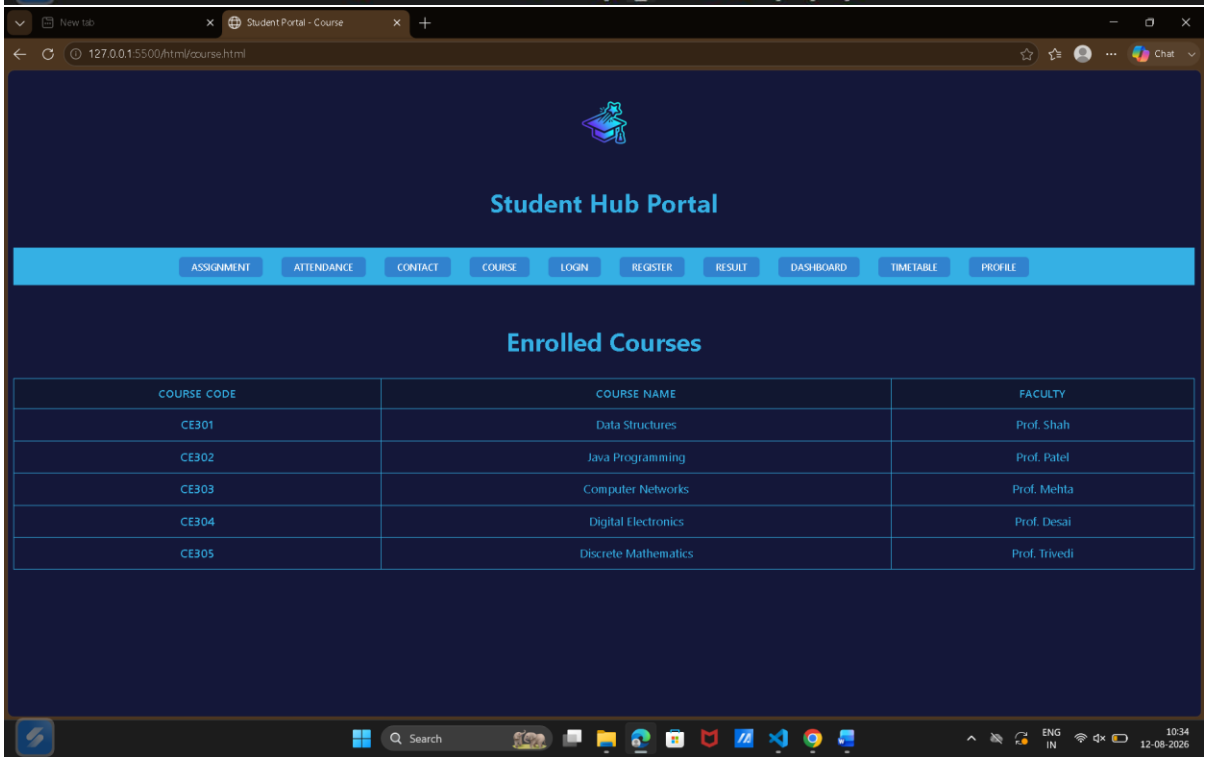

Full Name:
Email:
Password:
Gender:
Male
Female
Hobbies:
Reading
Sports
Music
City:
Ahmedabad
Address:
Submit
Reset


Student Registration
Full Name
Enter Full Name
Student ID
Enter Student ID
Email
Enter Email
Password
Enter Password
Register

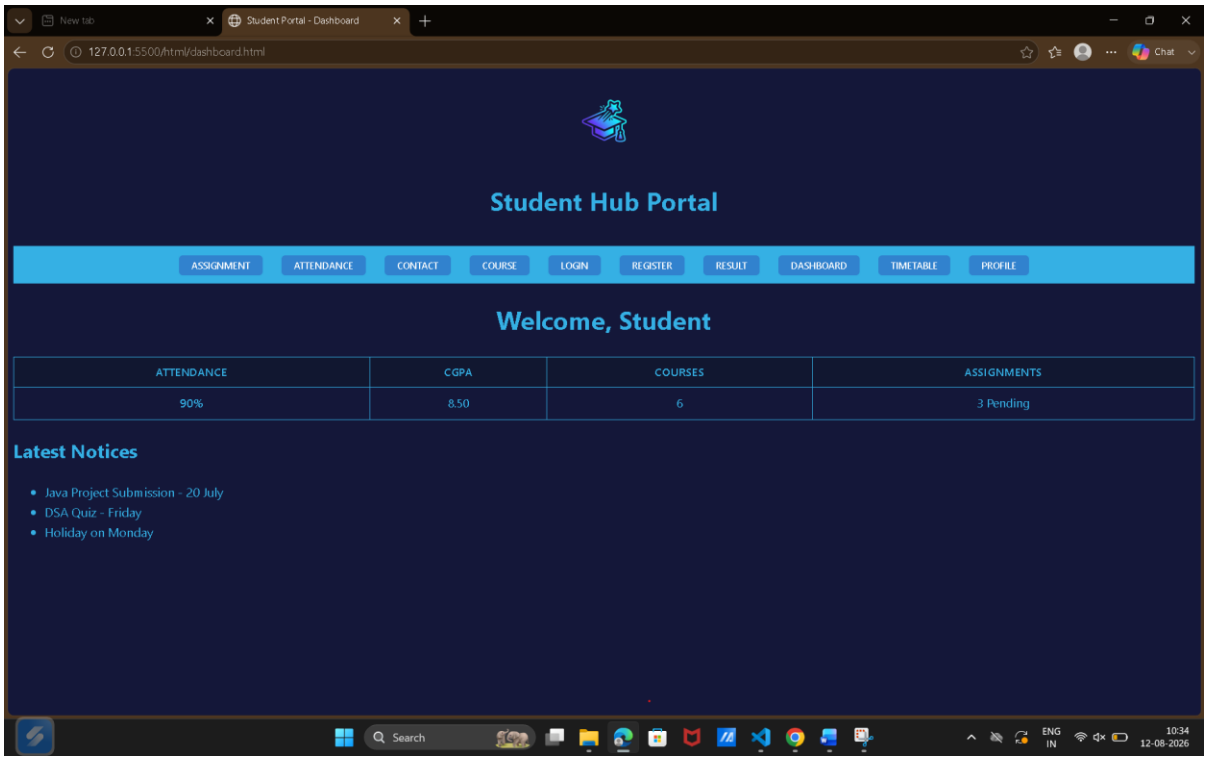

Student Portal - Assignment
127.0.0.1:5500/html/assignment.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Assignments
SUBJECT
ASSIGNMENT
DUE DATE
STATUS
Java Programming
Library Management System
20 July 2026
Pending
Data Structures
Linked List Program
22 July 2026
Completed
Computer Networks
OSI Model Report
25 July 2026
Pending
Discrete Mathematics
Graph Theory Worksheet
28 July 2026
Completed

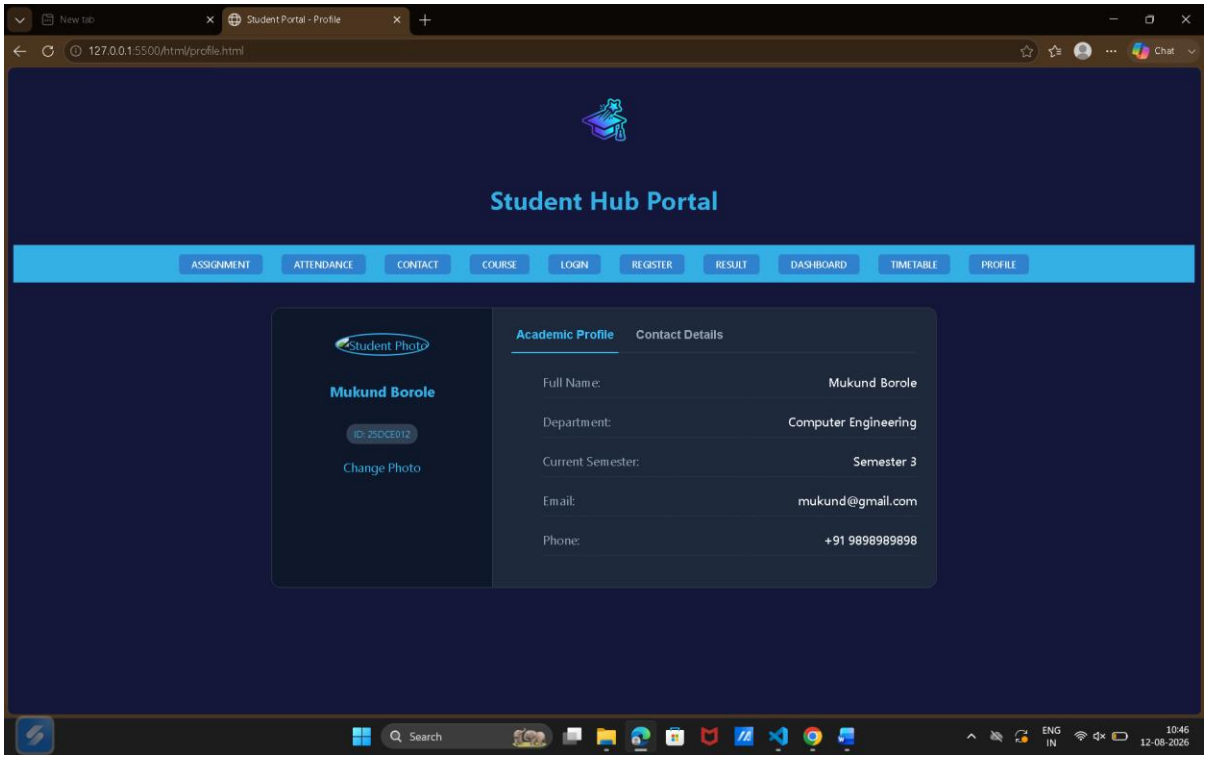

Student Portal - Attendance
127.0.0.1:5500/html/attendance.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Attendance Record
SUBJECT
CLASSES ATTENDED
TOTAL CLASSES
ATTENDANCE
Data Structures
27
30
90%
Java Programming
28
30
93%
Computer Networks
26
30
87%
Digital Electronics
29
30
97%
Discrete Mathematics
25
30
83%


Student Portal - Contact
127.0.0.1:5500/html/contact.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Contact Us
Name:
Enter Your Name
Email:
Enter Your Email
Message:
Enter Your Message
Send


Student Portal - Course
127.0.0.1:5500/html/course.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Enrolled Courses
COURSE CODE
COURSE NAME
FACULTY
CE301
Data Structures
Prof. Shah
CE302
Java Programming
Prof. Patel
CE303
Computer Networks
Prof. Mehta
CE304
Digital Electronics
Prof. Desai
CE305
Discrete Mathematics
Prof. Trivedi


Student Portal - Result
127.0.0.1:5500/html/result.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Semester Result
SUBJECT
GRADE
MARKS
Data Structures
A
87
Java Programming
A+
92
Computer Networks
B+
81
Digital Electronics
A
85
Discrete Mathematics
B
78
Overall CGPA : 8.50


Student Portal - Dashboard
127.0.0.1:5500/html/dashboard.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Welcome, Student
ATTENDANCE
CGPA
COURSES
ASSIGNMENTS
90%
8.50
6
3 Pending
Latest Notices
Java Project Submission - 20 July
DSA Quiz - Friday
Holiday on Monday


Student Portal - Profile
127.0.0.1:5500/html/profile.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Student Photo
Mukund Borole
ID: 25DCE012
Change Photo
Academic Profile
Contact Details
Full Name: Mukund Borole
Department: Computer Engineering
Current Semester: Semester 3
Email: mukund@gmail.com
Phone: +91 9898989898


Student Portal - Timetable
127.0.0.1:5500/html/timetable.html
Student Hub Portal
ASSIGNMENT
ATTENDANCE
CONTACT
COURSE
LOGIN
REGISTER
RESULT
DASHBOARD
TIMETABLE
PROFILE
Weekly Timetable
DAY | 9-10 AM | 10-11 AM | 11-12 PM | 1-2 PM | 2-3 PM
Monday | DSA | Java | CN | Maths | Lab
Tuesday | Java | DSA | DE | CN | Lab
Wednesday | Maths | CN | Java | DSA | Lab
Thursday | DE | Maths | Java | CN | Lab
Friday | DSA | Java | Maths | DE | Sports

The screenshot shows the StudentHub dashboard in mobile responsive mode. Content is rearranged into a smaller and easier-to-read layout suitable for a narrow screen.

D. GitHub Repository Link:

GitHub Repository: https://github.com/rehansaiyed72/student-hub-portal

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E. Challenges Faced:

During the implementation of the responsive StudentHub interface, the following challenges were faced:

1. Understanding how CSS Grid and Flexbox should be used for different layouts.
2. Making the same page work correctly on different screen sizes.
3. Adjusting the number of columns for cards and dashboard sections.
4. Preventing text, buttons, and navigation elements from overflowing on mobile screens.
5. Maintaining consistent spacing and typography across different devices.

6. Using media queries correctly for tablet and mobile breakpoints.
7. Keeping custom CSS organized without depending too much on inline styling.

These challenges were solved by testing the pages using the browser's responsive design mode and adjusting the Grid, Flexbox, spacing, and media-query rules accordingly.

F. Learning Outcomes:

After completing this practical, I learned:

1. How to create responsive web layouts using CSS3.
2. How to use Flexbox for arranging elements in one direction.
3. How to use CSS Grid for row and column-based layouts.
4. How media queries help websites adapt to different screen sizes.
5. How to follow a mobile-first approach when designing webpages.
6. How to maintain consistent typography, spacing, colors, and contrast.
7. How to test responsive webpages using browser developer tools.
8. How to organize CSS so that it can be reused across multiple pages.
9. How responsive design improves the usability of websites on different devices.

Faculty Remarks: